

S.T.A.R. Claim/Evidence Registry

Version 0.1 — Initial Controlled Baseline

Governed by S.T.A.R. Research Charter v0.2

1. Purpose

The S.T.A.R. Claim/Evidence Registry is the authoritative record of substantive mathematical, computational, empirical, theoretical, and speculative claims made within the S.T.A.R. research program.

The Registry exists to answer:

What does S.T.A.R. claim, what evidence supports the claim, what remains unverified, and what would falsify it?

The Registry is controlled by:

S.T.A.R. Research Charter v0.2

If a historical document conflicts with the epistemic standards of the Charter, the historical statement is preserved but its Registry classification is changed accordingly.

The Registry therefore separates:

[
 \text{Historical Claim}
]

from

[
 \text{Current Scientific Status}.
]

2. Registry Governing Rules

Rule 1 — A claim is not evidence

The fact that a proposition appears in an S.T.A.R. document does not constitute evidence supporting that proposition.

Rule 2 — A computation is not automatically an empirical validation

A computational result may establish:

- that code executed;
- that a mathematical quantity was calculated;
- that a numerical relationship occurred in a dataset;
- that a model fit a dataset.

It does not automatically establish:

- causation;
 - physical reality;
 - generalization;
 - predictive power;
 - mathematical theoremhood.
-

Rule 3 — Historical claims are preserved

Older claims shall not be silently deleted.

Instead they receive one of the following statuses:

- Active;
- Supported;
- Preliminary;
- Exploratory;
- Unverified;
- Reclassified;
- Falsified;

- Superseded;
 - Open.
-

Rule 4 — The strongest version of a claim is not automatically retained

When a source contains both a cautious and an extremely strong formulation, the Registry records the actual historical formulation but evaluates it according to the Charter.

For example:

“Arithmetic structure correlates with cosmic topology”

and

“The universe is generated by arithmetic structure”

are separate claims.

Evidence for the first does not automatically support the second.

Rule 5 — Model-dependent thresholds are not canonical

Historical thresholds such as:

[
 $W_2 < 10^{-2}$
]

are recorded as exploratory/model-dependent unless independently calibrated.

Rule 6 — Statistical significance and predictive utility are separate

A claim may have statistical significance without meaningful predictive utility.

Both must therefore be recorded separately.

Rule 7 — Discovery and confirmation are separate

A model, mapping, equation, feature, threshold, or classifier discovered using a dataset cannot use the same dataset as its sole independent confirmation.

3. Standard Registry Fields

Every substantive claim shall use the following schema.

Field	Required	Description
Claim ID	Yes	Permanent unique identifier
Claim Statement	Yes	Precise statement being evaluated
Source Module	Yes	ACSC, GLMPCT, ECC, RTCH, Φ , etc.
Source Document	Yes	Exact originating document
Historical Location	Preferred	Chapter, section, appendix, lemma, etc.
Epistemic Category	Yes	D/E/M/P/H/C/T/S
Claim Type	Yes	Mathematical, computational, empirical, physical, methodological, philosophical
Mathematical Object	If applicable	Equation, mapping, invariant, field, topology, etc.
Assumptions	Yes	Assumptions required for the claim
Prediction	If applicable	What the claim predicts
Observable	If applicable	Quantity that can actually be measured
Dataset	If applicable	Data used or proposed

Population	If applicable	Population to which claim applies
Null Model	If applicable	Required null hypothesis
Alternative Models	If applicable	Competing mappings/models
Parameters	If applicable	Parameters used
Parameter-Locking Status	Yes	Fixed, fitted, exploratory, unknown
Statistical Method	If applicable	Test/model used
Effect Size	If applicable	Magnitude of observed relationship
Uncertainty	If applicable	Error/interval/variance
In-Sample Result	If applicable	Result on discovery/training data
Out-of-Sample Result	If applicable	Independent test result
Replication Status	Yes	None, partial, independent, replicated
Reproducibility Status	Yes	Unknown, partial, reproducible, independently reproduced
Supporting Evidence	Yes	Evidence currently available
Contradictory Evidence	Yes	Known counterevidence/failures
Falsification Condition	Yes	What would count against the claim
Current Status	Yes	Current Registry disposition
Required Next Test	Yes	Next scientific action
Dependencies	Yes	BSD, dataset, software, mapping, etc.

Confidence/Support Level	Yes	Registry evidence level
Last Tested	Preferred	Date/version of latest test
Code Reference	Preferred	Script/repository/experiment
Notes	Optional	Additional historical context

4. Controlled Vocabulary

4.1 Epistemic Category

D — Derived

Mathematically follows from definitions and stated assumptions.

E — Empirical

Supported by reproducible observation or computation.

M — Model

A constructed mathematical or phenomenological model.

P — Postulate

An assumption deliberately adopted as a foundation of a proposed theory.

H — Hypothesis

A proposition being tested.

C — Conjecture

A proposition lacking proof or sufficient empirical confirmation.

T — Testable Prediction

A consequence capable of independent testing.

S — Speculative Interpretation

A possible interpretation not presently supported enough to function as an empirical hypothesis.

5. Evidence Status Vocabulary

Status	Meaning
SUPPORTED	Evidence currently meets Charter requirements for the stated claim
PRELIMINARY	Some evidence exists but important validation remains
EXPLORATORY	Useful observation/model, but not yet confirmatory
UNVERIFIED	Claim is stated but supporting evidence is insufficient
RECLASSIFIED	Historical claim has been downgraded under Charter v0.2
FALSIFIED	Specific formulation contradicted by documented evidence
SUPERSEDED	Replaced by a more precise formulation
OPEN	Scientifically unresolved
ACTIVE TEST	Currently undergoing experimental evaluation

6. Support Levels

S0 — Statement Only

Claim appears in a source but has no adequate supporting evidence.

S1 — Mathematical/Computational Construction

Object or computation exists.

S2 — Preliminary Correspondence

Initial empirical relationship exists.

S3 — Reproducible Statistical Correspondence

Result survives appropriate statistical and null testing.

S4 — Independent Replication

Result survives independent datasets or independent analysis.

S5 — Predictive Utility

Result demonstrates meaningful out-of-sample predictive performance.

S6 — Physical Theory

A mathematically consistent theory survives empirical physical constraints and produces novel successful predictions.

No claim may skip levels merely because its theoretical interpretation is ambitious.

7. Initial Registry

ACSC — Core Structural Claims

ACSC-001

Claim: Arithmetic invariants of elliptic curves can be projected into a geometric representation that may be compared with cosmic structure.

Source Module: ACSC

Epistemic Category: H / M

Claim Type: Mathematical / methodological

Historical Construction:

$(\phi_E, \theta_E, z_E, s_E)$
]

with coordinates derived from arithmetic invariants.

A simplified representation is:

$$\left(\begin{array}{l} \log|\Delta_E|, \\ \log N_E, \\ \alpha_{r_E} \end{array} \right).$$

Current Status: ACTIVE TEST

Support Level: S1

Assessment: This is a legitimate candidate mathematical construction. Its physical significance remains unestablished.

Required Next Test: Compare the mapping against null and alternative mappings under locked parameters.

Falsification: No reproducible correspondence beyond appropriate null expectations.

ACSC-002

Claim: The arithmetic projection ($\Phi(E)$) corresponds to the topology of large-scale cosmic matter distribution.

Epistemic Category: H

Claim Type: Empirical / topological

Historical Evidence Claimed: Persistent-homology and Wasserstein comparisons.

Historical Threshold: The source uses:

$$\left[\begin{array}{l} W_2 < 0.01 \end{array} \right]$$

as a proposed criterion.

Current Status: RECLASSIFIED — EXPLORATORY

Support Level: S2 at most pending independent reconstruction.

Reason: The historical threshold is model-dependent under Charter v0.2.

Required Next Test:

[
H_0:
D_{\rm arith}
]

is no more similar to

[
D_{\rm cosmic}
]

than appropriately randomized or alternative mappings.

ACSC-003

Claim: Arithmetic projection reproduces cosmic connectedness, filamentation, and void structure.

Epistemic Category: H

Historical Claim: The ACSC monograph describes these structures as having been reproduced by arithmetic projection.

Current Status: RECLASSIFIED — PRELIMINARY/EXPLORATORY

Support Level: S2

Critical Issue: The historical treatment identifies (β_1) directly with filaments.

Current Standard: Filament identification must primarily use geometric, skeleton, or graph-based methods. Persistent homology provides supporting topological characterization.

Required Next Test: Independent cosmic-web filament extraction followed by blind comparison with arithmetic topology.

ACSC-004

Claim: Arithmetic projection persistence diagrams align with SDSS/simulation topology at greater than 98% fidelity.

Epistemic Category: E / historical model criterion

Current Status: RECLASSIFIED — EXPLORATORY

Support Level: S1/S2 pending replication

Reason: The source provides the threshold as a validity criterion, but the Charter does not accept the historical percentage as a canonical acceptance standard.

Required Next Test: Define the fidelity metric, establish null distributions, lock parameters, and test on held-out data.

ACSC-005

Claim: Increasing the arithmetic dataset improves convergence toward cosmic topology.

Historical Mathematical Form:

```
[
W_2
\left(
D_k^{\rm arith},
D_k^{\rm cosmic}
\right)
\rightarrow 0.
]
```

Epistemic Category: C / H

Current Status: OPEN

Support Level: S1

Required Next Test: Learning-curve experiment with independent arithmetic samples and fixed cosmic target.

Falsification: No systematic improvement after controlling for density and normalization effects.

ACSC-006

Claim: The arithmetic projection converges to the cosmic manifold in the Gromov–Hausdorff sense.

Historical Form:

[
M_n
\rightarrow
M_{\rm cosmo}
]

in Gromov–Hausdorff distance.

Epistemic Category: C

Current Status: RECLASSIFIED — UNPROVEN CONJECTURE

Support Level: S0/S1

Reason: The source presents this as a theorem-like statement but the required mathematical assumptions and proof are not established by the documented computational evidence.

Required Next Test: Formal theorem development with precise spaces, metrics, compactness/completeness assumptions, and independently justified convergence.

ACSC-007

Claim: Rank determines the vertical/elevation coordinate of the projection.

Historical Form:

[
z=200r.
]

Epistemic Category: M

Current Status: EXPLORATORY MODEL

Support Level: S1

Interpretation: The factor 200 is a historical phenomenological scaling.

Required Next Test: Compare against alternative scaling functions and blind validation.

ACSC-008

Claim: Rank corresponds to cosmic dimensionality.

Historical correspondence:

[
r=0 $\rightarrow$ \text{flat/void-like structure},
]

[
r=1 $\rightarrow$ \text{linear/filament-like structure},
]

[
r=2 $\rightarrow$ \text{surface/cluster-like structure},
]

[
r=3 $\rightarrow$ \text{3-dimensional/supercluster structure}.
]

Epistemic Category: H**Current Status:** ACTIVE TEST**Support Level:** S0/S1**Required Next Test:** Directly test

[
P(r|mid C,F,V)
]

against appropriate environmental labels.

ACSC-009**Claim:** The regulator functions as a symbolic curvature/deformation variable.**Epistemic Category:** H / M**Current Status:** ACTIVE TEST**Support Level:** S1**Required Next Test:** Compare regulator-derived topography against independently measured density, lensing, gravitational potential, morphology, and SFR.

ACSC-010

Claim: Torsion can represent a hidden/dark structural component.

Historical Form:

[
$$D(E) = \log(1+T).$$

]

Epistemic Category: S / H

Current Status: RECLASSIFIED — SPECULATIVE

Support Level: S0

Required Next Test: Establish an observable prediction distinguishing torsion-based models from standard astrophysical explanations.

ACSC-011

Claim: Rank growth produces an analogue of dark-energy-driven expansion.

Historical Form:

[
$$z(t) = 200r(t).$$

]

Epistemic Category: S / H

Current Status: RECLASSIFIED — SPECULATIVE

Support Level: S0

Required Next Test: Develop a dynamical theory before testing cosmological expansion.

ACSC-012

Claim: Arithmetic invariants can replace the energy-momentum tensor or Einstein tensor as the fundamental description of gravity.

Epistemic Category: S

Current Status: RECLASSIFIED — SPECULATIVE

Support Level: S0

Required Conditions: A consistent field theory, action, conservation laws, known limits, stability, and novel observational predictions.

ACSC-013

Claim: A symbolic gravitational field can be defined from arithmetic quantities.

Historical construction:

[
 $\mathrm{G}(x)=\nabla s(x),$
 $\quad$
 $s=\log(1+\mathrm{R}).$
]

Epistemic Category: M / H

Current Status: ACTIVE THEORETICAL MODEL

Support Level: S1

Important distinction: Existence of the mathematical gradient does not establish equivalence with gravity.

ACSC-014

Claim: Symbolic energy can be constructed from Heegner height and conductor.

Historical form:

$\frac{\hat{h}(P_K)}{\log N}.$
]

Epistemic Category: M / H

Current Status: EXPLORATORY

Support Level: S1

Required Test: Derive conservation and dimensional interpretation independently before physical interpretation.

ACSC-015

Claim: Symbolic entropy can be represented using Mordell–Weil pairing data.

Historical form:

$$\log \det \left(\langle P_i, P_j \rangle \right).$$

Epistemic Category: M / H

Current Status: ACTIVE THEORETICAL TEST

Support Level: S1

Required Test: Mathematical domain/positivity/normalization analysis and independent thermodynamic interpretation.

ACSC-016

Claim: Symbolic entropy obeys a second-law-like monotonicity:

$$\mathcal{S}(E_n) \leq \mathcal{S}(E_{n+1}).$$

Epistemic Category: H / C

Current Status: OPEN

Support Level: S0

Required Test: Define the admissible arithmetic evolution and test monotonicity without selecting the sequence after observing the entropy values.

8. GLMPCT Claims

GLMPCT-001

Claim: A globally curved cosmic manifold can appear locally flat through an appropriate mapping/conformal representation.

Epistemic Category: M

Current Status: ACTIVE MODEL

Support Level: S1

Scientific Role: Provides the geometric/cartographic layer of GLMPCT.

Required Test: Specify the manifold, metric, conformal map, domain, and observable consequences.

GLMPCT-002

Claim: The universe can be modeled globally as a 3-sphere while maintaining locally approximately flat geometry.

Epistemic Category: H / M

Current Status: OPEN

Support Level: S0/S1

Required Test: Compare predicted global topology/curvature signatures with cosmological observations.

GLMPCT-003

Claim: Elliptic-curve rank corresponds to topological dimensionality of cosmic structures.

Epistemic Category: H

Current Status: ACTIVE TEST

Support Level: S0/S1

Required Test: Independent rank/environment classification.

GLMPCT-004

Claim: A rank-3 elliptic curve can serve as an analogue of a three-dimensional cosmic node.

Historical Evidence: A rank-3 curve was identified in the documented computational work.

Epistemic Category: M / H

Current Status: PRELIMINARY

Support Level: S1

Critical Limitation: The source itself identifies scarcity of diverse rank-3 curves as a limitation to generalizability.

Required Test: Larger rank-3 sample with independently defined astronomical targets.

GLMPCT-005

Claim: Virgo provides a calibration anchor for the physical scale of the arithmetic mapping.

Epistemic Category: M / E

Current Status: PRELIMINARY CALIBRATION

Support Level: S2 for the calibration experiment; not evidence of the full theory.

Required Test: Independent astronomical anchors.

GLMPCT-006

Claim: A scaled regulator can reproduce a target physical density/height.

Epistemic Category: E / M

Current Status: PRELIMINARY

Support Level: S2

Required Test: Blind prediction on new clusters rather than recalibration.

GLMPCT-007

Claim: Fibonacci/Lucas structures provide the correct recursive arithmetic grammar for cosmic mapping.

Epistemic Category: H / M

Current Status: EXPLORATORY

Support Level: S1

Required Test: Compare Fibonacci/Lucas against prime, random recurrence, and other mathematically motivated sequence families under locked criteria.

GLMPCT-008

Claim: The modified Strong BSD formula involving Fibonacci, (π) , and (ϕ) is supported by the cosmological mapping.

Epistemic Category: H

Current Status: FALSIFIED FOR TESTED FORMULATION

Reason: The appendices document failure of the modified Strong BSD formula on the tested curves.

Scientific Significance: This is an important negative result.

Registry Action: Preserve as evidence against arbitrary insertion of Fibonacci/ (π) /golden-ratio factors into established BSD relations.

GLMPCT-009

Claim: The failure of the modified BSD expression demonstrates a structurally meaningful GLMPCT–BSD connection.

Epistemic Category: C / H

Current Status: RECLASSIFIED — UNVERIFIED

Reason: Failure of a perturbed equation establishes that the perturbation fails; it does not by itself establish the cosmological interpretation.

9. Alternative Mapping Claims

MAP-001

Claim: The primary projection should not be required to be a strict bijection.

Epistemic Category: M

Current Status: ACTIVE FOUNDATIONAL DESIGN PRINCIPLE

Support Level: S1

Reason: The alternative-mapping document explicitly defines ($\backslash\Phi$) as a projection and allows many-to-one structure.

MAP-002

Claim: Multiple arithmetic curves may map to the same or equivalent projected cosmic node.

Epistemic Category: H / M

Current Status: ACTIVE MODEL

Support Level: S1

Required Test: Quantify equivalence classes/fibers and determine whether they possess predictive meaning.

MAP-003

Claim: Fibonacci/Lucas/ ϕ are necessary for the projection.

Historical Status: Not supported as necessary.

Current Category: M

Current Status: RECLASSIFIED — NON-ESSENTIAL

Reason: The alternative-mapping document itself states that alternative mappings do not require these constructions.

MAP-004

Claim: Fibonacci/Lucas/ ϕ define a useful self-similar subclass of arithmetic mappings.

Epistemic Category: M / H

Current Status: EXPLORATORY

Required Test: Out-of-sample comparison with alternative sequence families.

MAP-005

Claim: BSD is not required for the projection framework.

Epistemic Category: M / methodological claim

Current Status: SUPPORTED AS DESIGN REQUIREMENT

Support Level: S1

Interpretation: The mapping can be formulated using computable arithmetic invariants without assuming BSD.

MAP-006

Claim: Alternative mappings MCJ, PTD, and FT can provide BSD-independent controls.

Epistemic Category: M / T

Current Status: ACTIVE EXPERIMENTAL DESIGN

Support Level: S1

Required Test: Implement all mappings and compare under identical evaluation criteria.

MAP-007

Claim: High-rank instability may reveal physically meaningful boundary behavior.

Epistemic Category: S

Current Status: SPECULATIVE

Support Level: S0

Required Test: First determine whether instability is mathematical, numerical, algorithmic, or physical.

10. ECC Claims

ECC-001

Claim: An arithmetic projection can induce a cohomological structure on a symbolic entropy manifold.

Epistemic Category: H / M

Current Status: ACTIVE THEORETICAL PROGRAM

Support Level: S1

ECC-002

Claim: Every admissible symbolic projection induces a unique persistent cohomology class.

Historical Form: ECC Axiom 1.

Epistemic Category: P

Current Status: RECLASSIFIED — THEORETICAL POSTULATE

Support Level: S0

Important: Calling it an axiom establishes it as an assumption of ECC, not as an empirical fact.

ECC-003

Claim: The relevant entropy curvature can be represented by a closed non-exact 2-form:

$$\frac{1}{2}\omega_{AB}dY^A\wedge dY^B,$$

with

$$\begin{aligned} &[\\ &d\omega=0, \\ &\quad \\ &[\omega]\neq 0. \\ &] \end{aligned}$$

Epistemic Category: M / D conditional on construction

Current Status: ACTIVE MATHEMATICAL CONSTRUCTION

Support Level: S1

Required Test: Specify the target manifold and construct an explicit nontrivial cohomology class.

ECC-004

Claim: Symbolic entropy flux

$$\int_{\Sigma}\omega$$

acts as an invariant associated with projection identity.

Epistemic Category: H / M

Current Status: OPEN

Support Level: S1

Required Test: Demonstrate invariance under the specified allowed transformations.

ECC-005

Claim: Nontrivial (H^2) features correspond to persistent symbolic structures.

Epistemic Category: H

Current Status: ACTIVE THEORETICAL TEST

Required Test: Establish the precise relationship between de Rham cohomology and persistent homology rather than treating them as interchangeable.

ECC-006

Claim: Cohomological regularization improves symbolic machine-learning performance.

Epistemic Category: H

Current Status: PRELIMINARY

Support Level: S2 claimed computationally; independent validation required.

Required Test: Controlled ablation:

[
 baseline
 $\quad$ vs. $\quad$
 $\text{ECC-constrained model}$.
]

ECC-007

Claim: ECC-derived machine-learning attractors represent cohomological fixed points.

Epistemic Category: H / S

Current Status: EXPLORATORY

Support Level: S1/S2

Required Test: Define “cohomological fixed point” mathematically and test against a null model of ordinary machine-learning attractors.

ECC-008

Claim: ECC has been empirically validated as a computational theorem.

Historical Status: Explicitly claimed by the ECC source.

Current Category: E/H

Current Status: RECLASSIFIED — UNVERIFIED

Support Level: S2 at most for individual computational results.

Reason: Computational performance does not establish a mathematical theorem.

ECC-009

Claim: ECC provides a general symbolic organization framework beyond cosmology.

Epistemic Category: C / S

Current Status: SPECULATIVE GENERALIZATION

Support Level: S0

Required Test: Independent demonstrations in non-cosmological domains using pre-specified criteria.

11. RTCH Claims

RTCH is treated as a candidate variational completion of the broader program.

RTCH-001

Claim: A variational theory can combine Einstein–Hilbert gravity with scalar, symbolic, topological/cohomological, and matter sectors.

Epistemic Category: M

Current Status: ACTIVE THEORETICAL CONSTRUCTION

Support Level: S1

Required Test: Complete action, variation, boundary terms, field equations, conservation identities, and stability analysis.

RTCH-002

Claim: A closed target-space 2-form can be pulled back to spacetime:

[
 $F_Q = \Phi^* \omega$.
]

Epistemic Category: D / M

Current Status: MATHEMATICAL CONSTRUCTION

Support Level: S1

Condition: Requires a defined smooth map (Φ) and differential form (ω).

RTCH-003

Claim: The pullback satisfies:

[
 $dF_Q = 0$.
]

Epistemic Category: D

Current Status: DERIVED

Condition: Follows from

$$\begin{bmatrix} F_Q = \Phi^* \omega \end{bmatrix}$$

and

$$\begin{bmatrix} d\omega = 0 \end{bmatrix}$$

under the standard pullback/exterior-derivative relationship.

RTCH-004

Claim: The global quantity

$$\int_{\Sigma_2} F_Q$$

can function as a topological charge.

Epistemic Category: M / H

Current Status: OPEN

Support Level: S1

Required Test: Establish invariance under the intended deformations and gauge/field transformations.

RTCH-005

Claim: The scalar

$$\frac{1}{4} F_{Q\mu\nu} F_Q^{\mu\nu}$$

can enter a conformal matter coupling.

Epistemic Category: M

Current Status: ACTIVE THEORETICAL MODEL

Support Level: S1

RTCH-006

Claim: Matter may couple through

$$A^2(\phi, \mathcal{M}, \mathcal{J}_Q) g_{\mu\nu}.$$

]

Epistemic Category: M

Current Status: ACTIVE THEORETICAL MODEL

Support Level: S1

Required Test: Equivalence-principle, Solar-System, pulsar, lensing, and cosmological constraints.

RTCH-007

Claim: The conformal coupling produces a fifth-force-like acceleration for massive matter.

Historical point-particle form:

$$-c^2 h^{\mu\nu} \nabla_\nu \ln A.$$

]

Epistemic Category: D conditional on the action/coupling

Current Status: THEORETICAL DERIVATION REQUIRING FULL BOOKKEEPING

Required Test: Re-derive from the complete action and verify conventions/signs.

RTCH-008

Claim: Radiation does not experience the same conformal fifth force because:

$$\left[\begin{array}{l} p=\epsilon/3 \\ \end{array} \right]$$

implies

$$\left[\begin{array}{l} \bar{T}=-\epsilon+3p=0. \\ \end{array} \right]$$

Epistemic Category: D conditional on the specified matter model

Current Status: DERIVED MODEL RESULT

Required Test: Confirm from the complete field equations and transformed stress tensor.

RTCH-009

Claim: The full effective matter stress contains an additional (Q)-dependent contribution.

Historical form:

$$\left[\begin{array}{l} \bar{T}_{\{\mu\nu\}} \\ + \\ T_{\{\mu\nu\}}^{\{(Q,m)\}}. \\ \end{array} \right]$$

Epistemic Category: D / M

Current Status: ACTIVE DERIVATION

Required Test: Full metric variation from the complete action.

RTCH-010

Claim: RTCH provides a physical theory capable of explaining cosmological observations.

Epistemic Category: S / H

Current Status: SPECULATIVE

Support Level: S0

Required Test: Complete theory plus constraint-first phenomenology.

12. Cross-Module Claims

CROSS-001

Claim: Arithmetic rank may be related to cosmic-web morphology.

Sources: ACSC, GLMPCT

Epistemic Category: H

Current Status: PRIMARY ACTIVE HYPOTHESIS

Priority: Very High

Required Experiments:

[
P(r | cluster),
 $\quad$
P(r | filament),
 $\quad$
P(r | void).
]

Control for:

- redshift;
 - sampling density;
 - galaxy mass;
 - environment;
 - survey geometry;
 - selection effects.
-

CROSS-002

Claim: Rank is more informative than other arithmetic invariants.

Epistemic Category: H

Current Status: PRIMARY ACTIVE HYPOTHESIS

Comparison Set:

[
r,
 Reg ,
N,
 $|\Delta|$,
T.
]

Required Test: Nested model comparison and out-of-sample performance.

CROSS-003

Claim: Arithmetic topography can predict astronomical density.

Epistemic Category: H

Current Status: ACTIVE TEST

Required Observable:

[
 $\rho_{\text{matter}}(\mathbf{x})$.
]

Candidate Arithmetic Field:

[
 $h_{\text{arith}}(\mathbf{x})$.
]

CROSS-004

Claim: Arithmetic topography may correspond to gravitational potential.

Epistemic Category: H

Current Status: OPEN

Required Test: Independent weak-lensing and/or dynamical comparison.

CROSS-005

Claim: Arithmetic structure may contain information about star formation rate.

Epistemic Category: H

Current Status: ACTIVE ASTROPHYSICAL LABORATORY

Required Test: Predictive SFR modeling with strict train/validation/test separation.

Important: SFR success would constitute astrophysical evidence for the specific relationship tested, not automatic proof of ACSC.

CROSS-006

Claim: Arithmetic/topological structure evolves in correspondence with cosmic redshift evolution.

Epistemic Category: H

Current Status: FUTURE ACTIVE TEST

Required Test:

[
D_{\rm arithmetic}(\tau)
\quad\text{vs.}\quad
D_{\rm cosmic}(z).
]

CROSS-007

Claim: A successful arithmetic-cosmic correspondence can lead to a physical field theory.

Epistemic Category: C / S

Current Status: FUTURE THEORETICAL PROGRAM

Required Condition: Empirical correspondence must precede physical identification.

CROSS-008

Claim: The Hubble–Planck tension can be explained by the S.T.A.R. framework.

Epistemic Category: H / S

Current Status: DOWNSTREAM OPEN HYPOTHESIS

Required Condition: A consistent dynamical cosmological model must exist first.

13. Negative and Corrective Evidence Registry

The Registry must explicitly preserve evidence that constrains or weakens the theory.

NEG-001 — Modified Strong BSD Failure

The appendices document failure of modified BSD expressions involving Fibonacci numbers, (π) , and (ϕ) .

Status: FALSIFIED FOR THE TESTED FORMULATION.

Importance: High.

Interpretation: This is evidence against arbitrary numerical modification of established arithmetic identities.

NEG-002 — High-Rank Computational Instability

The appendices document computational instability for high-complexity curves.

Status: EMPIRICAL COMPUTATIONAL LIMITATION.

Importance: High.

Interpretation: High-rank/high-conductor calculations require explicit algorithmic validation.

NEG-003 — Scarcity of Rank-3 Curves

The GLMPCT documentation identifies scarcity of diverse rank-3 examples.

Status: DATASET LIMITATION.

Importance: High.

Interpretation: Rank-3 cosmic interpretations currently have limited generalizability.

NEG-004 — Engineered Feature Failure

The documented machine-learning work includes failures caused by infinities/extreme values in engineered features.

Status: COMPUTATIONAL FAILURE / DIAGNOSTIC EVIDENCE.

Importance: Moderate to High.

Interpretation: Feature engineering requires numerical controls before model performance can be interpreted.

14. Claim Dependency Graph

The Registry should eventually represent dependencies explicitly.

A simplified current graph is:

```
[  
  \boxed{  
    \text{Arithmetic Invariants}  
  }  
]
```

[
\downarrow
]

[
\boxed{
\Phi
}
]

[
\downarrow
]

[
\boxed{
\text{Arithmetic Geometry}
}
]

[
\downarrow
]

[
\boxed{
\text{Arithmetic Topography}
}
]

[
\downarrow
]

[
\boxed{
\text{Persistent/Geometric Topology}
}
]

[
\downarrow
]

[
\boxed{

$\text{Cosmic-Web Correspondence}$

$\downarrow$

$\boxed{\text{Rank/SFR/Physical Tests}}$

$\downarrow$

$\boxed{\text{Dynamics}}$

$\downarrow$

$\boxed{\text{Physical Theory}}$

Claims at a lower level must not be represented as proven merely because a higher-level interpretation exists.

15. Current Priority Classes

The initial Registry identifies four priority groups.

Priority A — Immediate Experimental Claims

These should receive the next experimental effort.

1. **ACSC-008** — rank/cosmic dimensionality
 2. **CROSS-001** — rank/cosmic-web morphology
 3. **CROSS-002** — rank versus other invariants
 4. **ACSC-002** — arithmetic/cosmic topological correspondence
 5. **MAP-006** — alternative mapping controls
 6. **CROSS-005** — arithmetic/SFR relationship
-

Priority B — Mathematical Validation

1. **ACSC-005** — convergence
 2. **ACSC-006** — Gromov–Hausdorff claim
 3. **ECC-003** — non-exact cohomology construction
 4. **ECC-004** — entropy flux invariant
 5. **RTCH-002** — pullback construction
 6. **RTCH-003** — closedness
 7. **RTCH-009** — metric variation
-

Priority C — Theoretical Development

1. **ACSC-013** — symbolic gravitational field
 2. **ACSC-014** — symbolic energy
 3. **ACSC-015** — symbolic entropy
 4. **ACSC-016** — entropy monotonicity
 5. **ECC-007** — cohomological attractors
 6. **RTCH-005/006** — (Q)-dependent matter coupling
-

Priority D — Speculative/Future

1. **ACSC-010** — torsion/dark matter
2. **ACSC-011** — rank/dark energy
3. **ACSC-012** — replacement of Einstein tensor
4. **RTCH-010** — complete physical theory
5. **CROSS-008** — Hubble–Planck tension

These claims remain part of the research program but should not drive the interpretation of earlier experiments.

16. Current Evidence Landscape

The initial audit suggests the following overall structure.

Mathematically defined

The strongest current material is concentrated in:

- arithmetic invariants;
- candidate projection maps;
- persistent-homology machinery;
- differential-form/cohomology constructions;
- candidate variational structures.

Computationally demonstrated

There are documented computational experiments involving:

- elliptic-curve generation;
- rank computation;
- regulator calculations;
- projection construction;
- persistent-homology pipelines;
- machine learning;
- symbolic regression;
- GLMPCT calibration;
- perturbation tests.

These establish computational activity and specific numerical results, but not automatically the broader physical claims.

Empirically interesting

The strongest experimental directions are:

- arithmetic/cosmic topology comparison;
- rank/environment relationships;
- arithmetic/SFR prediction;
- alternative mapping comparisons;

- independent astronomical validation.

Not yet established

The Registry currently treats the following as unestablished:

- arithmetic origin of cosmic structure;
 - rank as physical time;
 - rank as gravity;
 - torsion as dark matter;
 - rank growth as dark energy;
 - arithmetic replacement of Einstein gravity;
 - ECC as a theorem;
 - Gromov–Hausdorff convergence to the universe;
 - complete RTCH physical validity;
 - resolution of the Hubble–Planck tension.
-

17. Registry Decision Logic

For every new result, the following sequence shall be applied:

```
[
\boxed{
\text{Claim}
\rightarrow
\text{Definition}
\rightarrow
\text{Assumptions}
\rightarrow
\text{Prediction}
\rightarrow
\text{Null}
\rightarrow
\text{Test}
\rightarrow
\text{Replication}
\rightarrow
\text{Status}
}
]
```

If the claim is mathematical:

```
[
\text{Definition}
\rightarrow
\text{Proof}
\rightarrow
\text{Counterexample Search}.
]
```

If computational:

```
[
\text{Code}
\rightarrow
\text{Reproduction}
\rightarrow
\text{Independent Reproduction}.
]
```

If empirical:

```
[
\text{Dataset}
\rightarrow
\text{Null}
\rightarrow
\text{Statistical Test}
\rightarrow
\text{Out-of-Sample Test}.
]
```

If physical:

```
[
\text{Mathematical Consistency}
\rightarrow
\text{Known Limits}
\rightarrow
\text{Experimental Constraints}
\rightarrow
\text{Novel Prediction}.
]
```

18. Version Control

Each Registry revision shall include:

- Registry version;
- Charter version;
- source-document versions;
- date;
- new claims;
- modified claims;
- reclassified claims;
- falsified claims;
- newly replicated claims;
- newly unresolved claims.

A claim ID should never be reused for a different proposition.

If a claim changes substantially, create a new Claim ID and mark the previous claim:

SUPERSEDED.

19. Initial Baseline Rule

This Version 0.1 Registry establishes the following baseline:

Historical S.T.A.R. claims remain part of the research record, but their scientific status is determined by Research Charter v0.2 rather than by the language used in the original document.

This means that phrases such as:

- “proved”;
- “validated”;
- “theorem”;
- “law”;
- “definitive”;
- “confirmed”;
- “cosmos is”;
- “gravity is”;
- “dark matter is”;

must be interpreted according to their Registry classification.

A historical statement may remain quoted as historical evidence of what the project previously proposed without being treated as a current scientific conclusion.

20. Canonical Registry Principle

The Registry operates under the same governing principle as the Charter:

```
[
\boxed{
\textbf{The existence of a claim is not evidence for the claim.}
}
]
```

And the central operational rule is:

```
[
\boxed{
\text{Fixed construction}
\rightarrow
\text{Independent prediction}
\rightarrow
\text{Independent observation}
}
]
```

The Registry exists to make that chain visible.

21. Initial Registry Summary

At this initial stage, the S.T.A.R. program should be understood as containing three distinct layers.

Established mathematical/computational foundation

```
[
\boxed{
```

$$\begin{aligned}
 &\text{Arithmetic invariants} \\
 &+ \\
 &\text{candidate mappings} \\
 &+ \\
 &\text{topological machinery} \\
 &+ \\
 &\text{computational infrastructure} \\
 &\} \\
 &]
 \end{aligned}$$

Active scientific hypotheses

$$\begin{aligned}
 &[\\
 &\boxed{ \\
 &\text{Rank} \\
 &\leftrightharpoons \\
 &\text{cosmic structure} \\
 & } \\
 &]
 \end{aligned}$$

$$\begin{aligned}
 &[\\
 &\boxed{ \\
 &\text{Arithmetic topology} \\
 &\leftrightharpoons \\
 &\text{cosmic topology} \\
 & } \\
 &]
 \end{aligned}$$

$$\begin{aligned}
 &[\\
 &\boxed{ \\
 &\text{Arithmetic structure} \\
 &\leftrightharpoons \\
 &\text{SFR/galaxy evolution} \\
 & } \\
 &]
 \end{aligned}$$

Future theoretical possibilities

$$\begin{aligned}
 &[\\
 &\boxed{ \\
 &\text{Arithmetic} \\
 &\rightarrow \\
 &\text{Fields} \\
 &\rightarrow \\
 &\text{Dynamics} \\
 & } \\
 &]
 \end{aligned}$$

\rightarrow
\text{Gravity/Cosmology}
}
]

The research program should progress through these layers in order.

22. Registry Closing Statement

The Claim/Evidence Registry is not intended to determine in advance whether ACSC, GLMPCT, ECC, RTCH, or the broader S.T.A.R. framework is correct.

Its purpose is more fundamental:

To ensure that every claim has a known epistemic status, every piece of evidence has a defined role, every prediction has a falsification condition, and every theoretical interpretation remains distinguishable from the observations that motivate it.

Accordingly:

[
\boxed{
\textbf{S.T.A.R. preserves every claim.}
}
]

[
\boxed{
\textbf{S.T.A.R. tests every testable claim.}
}
]

[
\boxed{
\textbf{S.T.A.R. distinguishes models from evidence.}
}
]

[
\boxed{
\textbf{S.T.A.R. allows negative results to remain scientific results.}
}

```

}
]

[
\boxed{
\textbf{The evidence determines the status of the theory.}
}
]

```

Registry Initialization Record

Registry: S.T.A.R. Claim/Evidence Registry

Version: 0.1

Controlling Charter: S.T.A.R. Research Charter v0.2

Source modules initially audited:

- ACSC Monograph
- Arithmetic Invariants and Cosmological Geometry in Cartography
- Appendices for Arithmetic Invariants and Cosmological Geometry in Cartography
- Entropy Cohomology Conjecture
- Criteria for Alternative Mappings (Φ)

Initial registry state: ACTIVE / RECLASSIFICATION BASELINE

Next controlled operation:

```

[
\boxed{
\text{Claim Registry}
\rightarrow
\text{Experiment Registry}
\rightarrow
\text{Parameter Registry}
\rightarrow
\text{Mapping Registry}
}
]

```

The Claim/Evidence Registry is now the controlling index through which subsequent S.T.A.R. experiments should be interpreted.